

Henri Stewart Wilks-Stebbings

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Process-oriented Earth Scientist and analytical geomorphologist specialising in landslide dynamics, with expertise in sediment transport characteristics, hydrology, and process-modelling. Skilled in integrating field observations, remote sensing, and geospatial analysis to understand and model landscape evolution in tectonically-active and glaciated regions. Motivated to address the impact of climate change and uplift on geomorphological processes through innovative research and applied GIS solutions.

Education

PhD (dr.rer.nat) in Geology

University of Salzburg, Austria

March 2029

M.Sc. in Earth Sciences

Vrije Universiteit Amsterdam, Netherlands

GPA: 7.6

June 2025

Thesis (27 EC): *Constraining Landslide Rates, Controls, and Spatial Distributions in Calabria, Southern Italy*

Research Project (24 EC): *Landslide Frequency and Magnitude in the Southern Alps, New Zealand*

B.Sc. in Physical Geography (Honours)

Liverpool Hope University, England

GPA: Summa Cum Laude

June 2023

Dissertation (30 EC): *The Implications of Climate Change on High Mountain Geomorphology and Paraglacial Slope Failure*

Experience

Phd Researcher

University of Salzburg, Division of Geology and Physical Geography

Salzburg, Austria

Mar. 2029

Research Assistant, Geomorphology and Quaternary Sedimentology Group

Vrije Universiteit Amsterdam, Department of Earth Science

Amsterdam, Netherlands

Aug. 2024 – Apr. 2025

- Leading remote-sensing data acquisition for post-glacial landslide research in New Zealand
- Collaborating within a global, multi-disciplinary team of scientists (New Zealand, UK, Netherlands)
- Employing Z shell scripting and QGIS to processes geospatial raster data

Skills

Fieldwork Expertise: Geomorphological and geological mapping, field validation, sediment sampling, core logging

Technical Proficiency: QGIS, Python, Google Earth/Google Earth Engine, MATLAB, HBV-light, LandLab, Z shell

Analytical Skill: DEM analysis, hydrological modelling, statistics, and data visualisation

Communication: Technical report writing, scientific presentations, public speaking

Projects

Constraining Landslide Rates, Controls, and Spatial Distributions in Calabria, Southern Italy

(MSc Thesis, 27 EC)

Jun. 2025

- Student collaborator through MSc Thesis on the RELICT project: examining how sediment dynamics and inherited relief impact landscape response to tectonic perturbations
- My role in this project involves creating multi-year landslide inventories which help refine hazard assessment and landslide erosion rate estimates

Landslide Frequency and Magnitude in the Southern Alps, New Zealand

(MSc Research Project, 24 EC)

Dec. 2024

- Independent 7-month research project delivering a novel multi-year landslide inventory with >2,700 landslides

- Implemented advanced geomorphological and numerical modelling techniques to analyse landslide frequency–magnitude trends
- Developed data acquisition workflows using Python to automate large-scale spatial analysis in QGIS
- Co-author of a talk presented at the American Geophysical Union Fall Meeting 2024 and of the future publication in preparation, contributing to understanding the role of landslides and landslide dams on setting valley widths and fluvial aggradation and incision

The Implications of Climate Change on High Geomorphology and Paraglacial Slope Failures
(MSc Thesis, 30 EC)

Jun. 2023

- Investigated the impact of climate on hillslope processes, analysing mass movement frequency–magnitude
- Assessed geological properties, slope mechanics, and climate forcing to identify key drivers of instability
- Conducted systematic literature review to synthesise current research on slope failure mechanisms

Modelling Sediment Dynamics and Stratigraphic Basin Architecture
(Course: *Source to Sink*)

Dec. 2024

- Comprehensive assignment utilising MATLAB to model sediment transport, deposition, and stratigraphic architectures
- Optimised a grain size fining model to analyse the effect of subsidence rates, sediment supply, and initial grain size on sediment dynamics
- Constructed a 2D stratigraphic profile to visualise the impact of tectonic rate changes on sediment deposition patterns

Geotectonic Influence on Geomorphological Processes in Malta
(Course: *International Fieldwork*)

Apr. 2023

- Performed field investigation and site analysis to study geomorphological processes
- Delivered a technical report integrating geological and geomorphological data for regional interpretation

Soil Erosion Modelling Project: Slovenia Municipalities
(Course: *Advanced Spatial Analyses*)

Dec. 2023

- Modelled soil erosion using the Revised Universal Soil Erosion Equation for Slovenian municipalities
- Developed automated workflows in Python utilising *rasterio*, *rioxarray*, and *geopandas* for spatial data analysis
- Created automated scripts to compute and visualise erosion

Rainfall-runoff Modelling Project: Kirchner Method
(Course: *Catchment Hydrology*)

Sep. 2024

- Built a hydrological model using the backward Euler method, simulating discharge for catchment-scale runoff
- Compared model predictions with observed data to evaluate accuracy and performance
- Applied principles of mass conservation and continuity equations to ensure realistic flow representation

HBV Hydrological Model: Dinkel Catchment
(Course: *Catchment Hydrology*)

Oct. 2024

- Derived rainfall inputs using Thiessen polygons for semi-distributed hydrological modelling in HBV-light
- Calibrated the model using manual and Monte Carlo simulations, employing statistical goodness-of-fit metrics
- Conducted flood frequency analysis with the Gumbel distribution and incorporated hydraulic conductivity into simulations

Remote Sensing Application: Engabreen Glacier, Northern Norway
(Course: *Environmental Remote Sensing*)

Jan. 2024

- Mapped glacier terminus extent using Landsat 7 imagery and the Normalised Difference Snow Index (NDSI)
- Analysed glacier area changes and mass balance dynamics via Google Earth Engine and JavaScript workflows
- Presented research findings at a poster session, highlighting remote sensing applications in glacial retreat monitoring

Paleosedimentology and Terrain Analysis

(Course: *Paleoclimate Change and Environmental Impacts*)

Apr. 2024

- Conducted fieldwork to collect sediment cores from terrestrial and marine environments
- Produced a technical report detailing core descriptions, sampling strategies, and sedimentological profiles
- Advanced laboratory techniques, including laser diffraction, thermogravimetry, magnetic susceptibility, and XRF for geochemical analysis
- Presented findings at a scientific poster session

Professional Development

Landslide Mapping Hackathon

May. 2025

- Organised and led a one-day landslide mapping event for undergraduate students, focusing on geomorphological analysis and spatial data interpretation
- Delivered hands-on instruction in landslide delineation techniques using satellite imagery and GIS tools
- Demonstrated methods for estimating landslide volumes using area-volume scaling relationships
- Served as a primary point of contact for participants, providing technical guidance and support throughout the event

Fugro, Engineering Workshop

Feb. 2025

- Interpreted seismic cross-sections and correlated with geotechnical data
- Geotechnical location planning
- Identified geohazards relevant to offshore wind farm development

Publications

Tectonic and climatic controls on valley-floor width along the Alpine Fault, New Zealand (Clubb, F.J., Roda-Boluda, D.C., Wilks-Stebbins, H. and de Vries, L., *American Geophysical Union Fall Meeting 2024*, oral presentation by F. Clubb.)

Awards

Awarded the **Liverpool Geological Society Prize** as Best Performing Liverpool Hope Student in Physical Geography, 2023.

Other Experience

Groundworks Site Manager

Joseph Davidson Groundworks & Construction

Bishop Auckland, England

May 2022 – Dec. 2022

Heavy Machinery Operator

Hambleton Farming Company

Northallerton, England

Mar. 2020 – Aug. 2020

Heavy Machinery Field Cultivation Operator

Scott Simpson Contracting Ltd

Oamaru, New Zealand

Jan. 2020 – Mar. 2020

Heavy Machinery Field Cultivation Operator

Chris Swanson Contracting: Cultivation, Excavation, and Drilling

Dunsandel, New Zealand

Sep. 2019 – Dec. 2019

References

Dr. Duna Roda-Boluda

Assistant Professor, Faculty of Science, Geology and Geochemistry
d.c.rodaboluda@vu.nl

Dr. Benjamin Campforts

Assistant Professor, Faculty of Science, Earth and Climate
b.campforts@vu.nl

Dr. Kevin Crawford

Senior Lecturer, School of Computer Science and the Environment
crawfok@hope.ac.uk

